

TWO-LOOP DIAGRAMS IN YANG-MILLS THEORY

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Abstract: A calculation of the renormalisation constants of the Yang-Mills field to $O(g^4)$ is presented. The function $\beta(g)$ is hence evaluated to $O(g^5)$ and possible implications for gauge theories of the strong interactions discussed.

1. Introduction

Great interest has been aroused recently in non-Abelian gauge theories of the strong interactions because of the observation [1] that such theories can exhibit free-field asymptotic behaviour at large Euclidean momenta, hence providing a natural field theoretic framework for the explanation of the scaling phenomena observed in electroproduction and neutrino-production experiments [2–4]. However it seems that the introduction of scalar mesons with non-vanishing vacuum expectation values to give masses to the gauge fields destroys the free field behaviour, because of the quartic interactions inevitably present in a renormalisable theory [2]. This observation led to the conjecture [2, 5] that the strong gauge symmetry is in fact exact, and that the absence of massless vector bosons is associated with the infra-red divergences involved in such a theory.

Asymptotic freedom for non-Abelian theories is a consequence of the behaviour near the origin of the calculable function $\beta(g)$; in view of the above conjecture its behaviour away from the origin is also of interest; a zero of β for some finite value of g would correspond to an infrared stable fixed point, hence rendering the conjecture suspect. In the absence of non-perturbative approaches a two-loop calculation of β is therefore of interest, and is performed here, using the dimensional regularisation technique of 't Hooft and Veltman [6].

We find (for a theory without fermions) that the second term in β has the same sign as the first, which suggests that β does not have another zero near the origin.

2. Calculation of the renormalisation constants

The Lagrangian, written in terms of renormalised quantities, is

$$L = -\frac{1}{4}Z_3(\partial_\mu W_\nu^a - \partial_\nu W_\mu^a) - \frac{1}{2\alpha}(\partial_\mu W_\nu^a)^2 - Z_1 g f_{abc} \partial_\mu W_\nu^a W_\mu^b W_\nu^c \\ - \frac{1}{4}g^2 \frac{Z_1^2}{Z_3} f_{abc} f_{ade} W_\mu^b W_\nu^c W_\mu^d W_\nu^e - Z_3^F \partial_\mu \phi_a^* \partial_\mu \phi_a - Z_1^F g f_{abc} \partial_\mu \phi_a^* W_\mu^b \phi_c \quad (1)$$

ϕ_a is the ghost field; α is the renormalised gauge parameter. (Note that the longitudinal part of the W propagator is unrenormalised.) The f_{abc} are the totally antisymmetric structure constants of the underlying gauge group G . They obey the relation

$$f_{acd} f_{bcd} = C_2(G) \delta_{ab} \quad (2)$$

where $C_2(G)$ is the value of the quadratic Casimir operator in the regular representation of G . All products of structure constants which occur in the Wick expansion (up to $O(g^4)$, at least) depend only on $C_2(G)$ and the tensors f_{abc} and δ_{ab} .

We use the metric conventions and Feynman rules as given by 't Hooft [7]; and since the Feynman integrals are to be performed in n dimensions it is convenient to define a new coupling constant u by

$$g = m^{\frac{1}{2}\epsilon} u \quad (3)$$

where $\epsilon = 4 - n$, and m is an arbitrary mass. Simple dimensional analysis now shows that u is dimensionless.

The subtraction terms $(Z_1 - 1)$, $(Z_3 - 1)$ etc. are functions of u , α and ϵ and are chosen (following 't Hooft [8]) to contain only inverse powers of ϵ in an expansion about $\epsilon = 0$.

We choose to work throughout in the renormalised Feynman gauge ($\alpha = 1$).

2.1. One-loop contributions

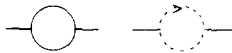


Fig. 1.



Fig. 2.

From the diagrams of fig. 1 we obtain

$$\Pi_{\mu\nu}^{(1)}(p) = i g^2 \pi^{\frac{1}{2}n} C_2(G) \delta_{ab} \Gamma(2 - \frac{1}{2}n) C(n) (p^2)^{\frac{1}{2}n-2} [p^2 \delta_{\mu\nu} - p_\mu p_\nu] \quad ,$$

ab

where

$$C(n) = \frac{1}{2} [(4n-8) B(\frac{1}{2}n, \frac{1}{2}n) + (6-n) B(\frac{1}{2}n-1, \frac{1}{2}n-1)] \quad (4)$$

(We have assumed that $\int d^n k / k^2 \equiv 0$; and when we come to two-loop diagrams we will also need $\int d^n k / (k^2)^{3-\frac{1}{2}n} = 0$. If we write

$$\int \frac{d^n k}{(k^2)^\alpha} = \int \frac{d^n k}{(k^2)^\alpha} \frac{(p-k)^2}{(p-k)^2} = p^2 \int \frac{d^n k}{(k^2)^\alpha (p-k)^2} - 2 \int \frac{p \cdot k}{(k^2)^\alpha (p-k)^2} \\ + \int \frac{d^n k}{(k^2)^{\alpha-1} (p-k)^2} ,$$

naive evaluation gives zero for general α . See, however, ref. [9] for a discussion of the mathematical niceties involved.)

Writing

$$g^2(p^2)^{\frac{1}{2}n-2} = u^2(p^2/m^2)^{\frac{1}{2}n-2} ,$$

and expanding about $n = 4$ we find

$$(Z_3 - 1)^{(1)} = \frac{10}{3} \frac{u^2}{16\pi^2} C_2(G) \frac{1}{\epsilon} . \quad (5)$$

Similarly from fig. 2 we find

$$(Z_3^F - 1)^{(1)} = \frac{u^2}{16\pi^2} C_2(G) \frac{1}{\epsilon} . \quad (6)$$

The vertex correction diagrams are evaluated with one external particle four-momentum set to zero. This makes the integrals easier and does not introduce any spurious singularities.

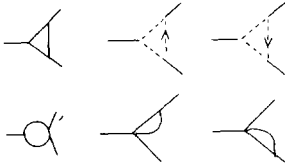


Fig. 3.

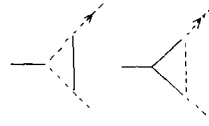


Fig. 4.

We find (see figs. 3 and 4)

$$(Z_1 - 1)^{(1)} = \frac{4}{3} \frac{u^2}{16\pi^2} C_2(G) \frac{1}{\epsilon} , \quad (Z_1^F - 1)^{(1)} = -\frac{u^2}{16\pi^2} C_2(G) \frac{1}{\epsilon} . \quad (7)$$

For these results in an arbitrary gauge see ref. (2), where a momentum cut-off is employed.

2.2. Two-loop contributions

We calculate Z_3 , Z_3^F and Z_1^F to $O(g^4)$, relying on the Slavnov-Taylor [10] identity

$$\frac{Z_1^F}{Z_3^F} = \frac{Z_1}{Z_3}, \quad (8)$$

to determine Z_1 .

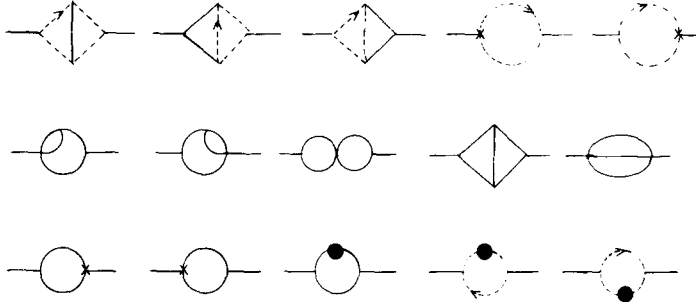


Fig. 5.

The $O(g^4)W$ self-energy diagrams are shown in fig. 5. (Crosses denote $O(g^2)$ contributions from the counter-terms (7); a blob on propagator or vertex denotes that it is renormalised to $O(g^2)$.)

Evaluation of the integrals is straightforward apart from those arising from diagrams with five internal lines. These are of the form

$$I_{\mu\nu}(p) = \int \frac{N_{\mu\nu}(p, k, q) d^n k d^n q}{k^2 q^2 (k+p)^2 (q+p)^2 (q-k)^2}, \quad (9)$$

where $N_{\mu\nu}$ is a fourth-order polynomial in p , k , and q .

Now any term in which one of the propagator denominators is missing is easily integrated, e.g.

$$\begin{aligned} \int \frac{d^n k d^n q}{k^2 q^2 (k+p)^2 (k-q)^2} &= i \pi^{\frac{1}{2}n} \Gamma(2 - \frac{1}{2}n) B(\frac{1}{2}n-1, \frac{1}{2}n-1) \int \frac{d^n k}{(k^2)^{3-\frac{1}{2}n} (p+k)^2} \\ &= \frac{-\pi^n \Gamma(2 - \frac{1}{2}n) \Gamma(4-n)}{\Gamma(3 - \frac{1}{2}n)} B(\frac{1}{2}n-1, \frac{1}{2}n-1) B(n-3, \frac{1}{2}n-1) (p^2)^{n-4}. \end{aligned}$$

Hence $N_{\mu\nu}$ can be reduced to three terms:

$$(i) N_1 = p^4 \delta_{\mu\nu}; \quad (ii) N_2 = p^2 k_\mu k_\nu; \quad (iii) N_3 = p^2 k_\mu q_\nu.$$

$(I_1)_{\mu\nu}$ is finite in the limit $\epsilon \rightarrow 0$ and can therefore be ignored. Let

$$(I_i)_{\mu\nu} = p^2 A_i \delta_{\mu\nu} + B_i p_\mu p_\nu, \quad \text{for } i = 2, 3 \quad (10)$$

The pole parts of A_i and B_i can be determined by multiplying both sides of (10) successively by $\delta_{\mu\nu}$ and p_μ and performing the integrations.

In the resulting expressions the Γ and B functions are expanded about $n = 4$, using

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + O(\epsilon),$$

where $\gamma = -\int_0^\infty e^{-t} \ln t \, dt$ is Euler's constant.

The result has the form

$$\Pi_{\mu\nu}^{(2)}(p) \sim \left[\pi^n \left\{ \frac{A}{\epsilon^2} - \frac{A\gamma}{\epsilon} + \frac{B}{\epsilon} \right\} \left(\frac{p^2}{m^2} \right)^{-\epsilon} - \pi^{\frac{1}{2}n} \left\{ \frac{A}{2\epsilon^2} - \frac{A\gamma}{\epsilon} + \frac{C}{\epsilon} \right\} \left(\frac{p^2}{m^2} \right)^{-\frac{1}{2}\epsilon} \right] \\ \times u^4 (p^2 \delta_{\mu\nu} - p_\mu p_\nu).$$

As required by renormalisability, there is no counter-term of the form $(1/\epsilon) \ln p^2$. Note that the dependence on γ also disappears; presumably because its occurrence is a feature of the particular regularisation scheme adopted.

The ghost self-energy and vertex contributions are calculated in similar fashion from figs. 6 and 7. (The vertex diagrams present no additional difficulties if as before one external momentum is put equal to zero.)

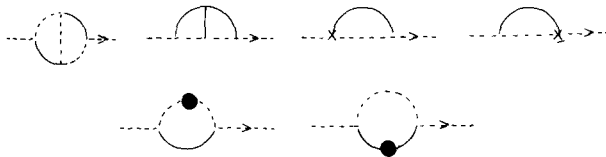


Fig. 6.

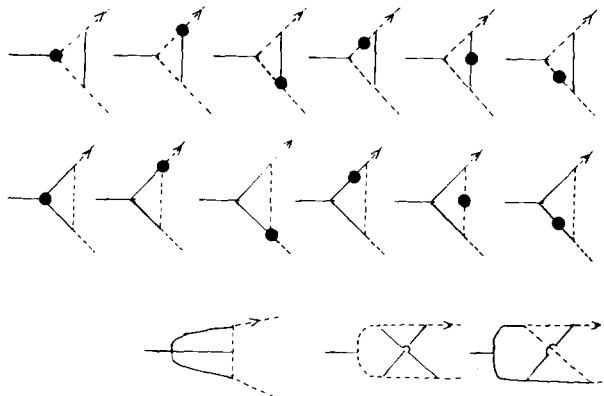


Fig. 7.

The results are

$$\begin{aligned}
 (Z_3 - 1)^{(2)} &= \frac{u^4 (C_2(G))^2}{2^8 \pi^4} \left[-\frac{25}{3} \frac{1}{\epsilon^2} + \frac{23}{4} \frac{1}{\epsilon} \right], \\
 (Z_3^F - 1)^{(2)} &= \frac{u^4 (C_2(G))^2}{2^8 \pi^4} \left[-\frac{4}{\epsilon^2} + \frac{49}{24} \frac{1}{\epsilon} \right], \\
 (Z_1^F - 1)^{(2)} &= \frac{u^4 (C_2(G))^2}{2^8 \pi^4} \left[\frac{5}{2} \frac{1}{\epsilon^2} - \frac{3}{4} \frac{1}{\epsilon} \right].
 \end{aligned} \tag{12}$$

3. Calculation of β

The renormalisation group equations for the N -particle single-particle irreducible Green functions $\Gamma^N(P_i, m, \alpha, u)$ are

$$\left[m \frac{\partial}{\partial m} + \beta(u) \frac{\partial}{\partial u} - N\gamma(u, \alpha) + \delta(u, \alpha) \frac{\partial}{\partial \alpha} \right] \Gamma^{(N)}(p_i, m, \alpha, u) = 0, \tag{13}$$

where

$$\beta(u) = m \left. \frac{\partial u}{\partial m} \right|_{g_0, \alpha_0 \text{ const.}}. \tag{14}$$

g_0, α_0 are the unrenormalised coupling constant and gauge parameters, given by

$$g_0 = \frac{Z_1(u, \alpha)}{Z_3^{\frac{1}{2}}(u, \alpha)} g, \tag{15}$$

$$\alpha_0 = Z_3 \alpha. \tag{16}$$

The solution of (13) can be expressed in the form

$$\Gamma^{(N)}(\lambda p_i, u, \alpha) \sim \Gamma^{(N)}(p_i, \bar{u}, \bar{\alpha}), \tag{17}$$

where

$$\frac{d \bar{u}}{d(\ln \lambda)} = \beta(\bar{u}). \tag{18}$$

(17) expresses the fact that rescaling the external momenta is equivalent to changing the coupling constant (and the gauge). Eq. (18) show that the zeros of β control the behaviour of the effective coupling constant in the high or low momentum limits.

The question of the gauge dependence of β now arises. Gross and Wilczek [2] prove (and verify directly) that the leading term in β is gauge independent but leave open the possibility of gauge dependence in higher order terms. We believe,

however, that β is gauge invariant to all orders, on the basis of the following argument*.

If we calculate the three-point function using the unrenormalised Lagrangian, the resulting amplitude $F(p, k, q, g_0, \alpha_0)$ must correspond to a gauge invariant S -matrix element. This implies that

$$F(p, k, q, g_0, \alpha_0) - F(p, k, q, g_0, \alpha'_0) = O(p^2, k^2, q^2), \quad (19)$$

which says simply that the difference between the amplitudes in different gauges vanishes on the mass shell. If we now expand F in powers of ϵ , the renormalised coupling constant $g(g_0, \alpha_0)$ is given by the terms in F which have poles at $\epsilon = 0$; but these have no dependence on the external momenta and so

$$\frac{\partial g}{\partial \alpha_0}(g_0, \alpha_0) = 0, \quad (20)$$

whence it follows that β does not depend on α .

From eqs. (3), (14) and (15) we find

$$\beta(u) = \frac{-\frac{1}{2}\epsilon}{\frac{d}{du} [u \ln Z_1/Z_3^{\frac{2}{3}}]}. \quad (21)$$

Substituting in (21) for Z_1 and Z_3 (using eq. (8)) we find that the term in $1/\epsilon^2$ cancels (see ref. [8]) and

$$\beta(u) = -\frac{11}{3}u^3 \frac{C_2(G)}{16\pi^2} - \frac{34}{3}u^5 \left(\frac{C_2(G)}{16\pi^2} \right)^2. \quad (22)$$

(Note that the contributions of Z_3 and Z_1^F/Z_3^F have the same sign for each term.)

Of course no definite conclusions can be drawn but the sign of the u^5 term does support to some extent the conjecture that the strong gauge group is unbroken.

Since this paper was initially submitted we have extended the result to include a multiplet of fermions. The full expression for $\beta(u)$ is then

$$\beta(u) = \frac{Au^3}{16\pi^2} + \frac{Bu^2}{(16\pi^2)^2}, \quad (23)$$

where

$$A = -\left[\frac{11}{3}C_2(G) - \frac{8}{3}T(R)\right], \quad (24)$$

and

$$B = \frac{20}{3}C_2(G)T(R) + 4C_2(R)T(R) - \frac{34}{3}C_2^2(G), \quad (25)$$

where

$$C_2(R)I = R^a R^a,$$

* I am particularly indebted to Dr. J.C. Taylor for conversations on this subject.

and

$$\delta^{ab}T(R) = \text{Tr} [R^a R^b] ,$$

where R^a are the generators of G in the fermion representation.

Thus in the model usually discussed ($G = \text{SU}(3)$, with three fermion triplets) we have $B \approx -70$, so the sign of B is not changed by the presence of the fermions. In the model of Bailin and Love (Sussex preprint January 1974) who take $G = \text{SU}(2)$ with three fermion triplets, we find $B \approx +80$, which renders dubious their conjecture that the model has an attractive fixed point near the origin.

A similar calculation (but performed in the Landau gauge) has been reported by Belavin and Migdal, (Landau preprint January 1974) who obtain $-\frac{183}{16}$ as the coefficient of the $C_2^2(G)$ term in eq. (25). As yet we have no explanation of this discrepancy.

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